

Got your Back- Bank Board Network in Emerging Markets

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Abstract

This paper study the nature of the bank board network in India and the relation between board network and financial stability of banks. Using a sample of listed public-sector and private banks in India, we find that the larger board network is related to improvements in financial stability given by asset quality, z-score of insolvency risk, and volatility of profits. Further, the board network is more important for the sub-set of public sector banks in India that are characterized by large and increasing non-performing assets. Our findings support the integrated resource dependence view of the board where well-connected boards are better able to monitor the management and provide strategic advice necessary to reduce risk. The findings of the study have practical implications related to the governance structure of bank boards in India. Unlike for non-financial firms, restricting the number of directorial positions for bank directors may not have any desirable firm outcome for banks.

Keywords: Board network, financial stability, asset quality, banks, India

JEL Codes: G21, G32, G34, G38

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1. Introduction

The banks are one of the primary intermediaries that mobilize household savings and allocate capital to the firms in the economy. The well-functioning banking industry helps in the efficient allocation of capital to various sectors of the economy and is related to economic growth. The global financial crisis of 2008 (GFC) exposed the vulnerabilities of the banking sector on account of the excessive risk-taking behavior of bankers. The banks are prone to over-extend credit during an economic upturn and misallocate capital by lending to bad borrowers. On the other hand, during the downturn as the value of the collateral falls, the bank lending contracts. The banks can over-amplify the business cycles in the economy. In response to the GFC, the Basel Committee on Banking Supervision came up with Basel III norms to improve the resilience of the banking sector to shocks. As banks play a central role in any economy, the good governance of banks is essential for maintaining public trust in the banks. The GFC once again highlighted the importance of good corporate governance practices of banks. The OECD Steering Group on Corporate Governance did suggest that failure of the boards of financial firms was one of the reasons that contributed towards the GFC (Kirkpatrick, 2009). The Walker Review (2009) also released a set of 39 specific recommendations related to board composition, board performance, institutional shareholders, risk governance, and remuneration for UK banks in the aftermath of the GFC.

The corporate governance of banks requires special attention, as banks are heavily regulated and are more opaque compared to non-financial firms (Levine, 2004). Further, as bank failure generates economy-wide negative externalities, large banks have an implicit guarantee against failure, and this can breed excessive risk-taking tendencies of managers of large banks (Haan and Vlahu, 2016). In general, the bank boards are larger than the boards of non-financial firms owing to the complexities of banking operations. Also, in many countries, the corporate governance regulations applicable to the financial and non-financial firms are different. Studies have highlighted that the board structure of banks (Adams and Mehran, 2012; Pathan and Faff, 2013), ownership structure (Laeven and Levine, 2009) affects bank outcomes. Elyasiani and Zhang (2015) do not find that busyness of bank directors adversely influences the risk-taking behavior of banks, instead multiple directors are related to the superior performance of banks. The finding suggests that multiple directorships of bank directors could be a proxy of the financial expertise of the bank director. More recently, the study by Adelbadie

and Salama (2019) find that board network reduces the credit, insolvency, and capital risks of banks in the United States. The review paper on corporate governance in banks by John et al. (2016) also emphasize that future research on the importance of corporate governance of banks need to further focus on the director network.

This study attempts to provide an understanding of the bank board network in India – an emerging market economy characterized by a bank-based financial system. The study provides valuable insights regarding the nature of the bank board network in India since 2013. We give an account of both direct and indirect connections of the bank boards. The direct connections of the bank are measured by the degree centrality of the board. The bank director by virtue of sittings on multiple boards will know several other directors. These directors form the direct connections of the bank director. We also capture the indirect connections of the bank directors by considering the second level connections. When the bank director occupies another board seat, the board members of the connected firm are also likely to know additional directors due to their multiple board seats. The direct connections of the bank director's first connections represent the indirect connections. In the last decade, the direct connections of banks in India have remained relatively stable at around 20 connections. On the other hand, the average number of indirect connections of the bank has increased from 120 to over 170 connections in the last decade. Also, the number of connections are higher for private banks than for public sector banks in India.

Further, the study also investigates the relation between bank board network and the financial stability of the Indian banks. Several studies have documented the relationship between director network and firm outcomes. Most of the research on the role of director networks focus on non-financial firms. We analyze whether direct as well as indirect bank board connections are related to the financial stability of the banks in India. We consider multiple financial stability measures - gross and net non-performing asset ratios, z-score of insolvency risk and volatility of ROA. Using a 2SLS IV method, we find that a large board network is related to the higher financial stability of banks. The results suggest that a larger board network improve the resources available to the bank. Having more resourceful and experienced directors help banks to improve financial stability through several channels. First, the more experienced networked director can help the bank to renegotiate with the borrowers in the debt restructuring process. Second, the networked director having access to a larger pool of information may help in reducing the information asymmetry between the borrower and the bank and, in turn, reduce the probability of extending loans to bad quality borrowers. Thirdly,

the connected director with vast experience provides strategic advice, which helps the banks to reduce over-time volatility in performance. We find that the board network plays a more important role in improving the asset quality of public sector banks (PSBs) in India. The asset quality of PSBs deteriorated significantly during our study period. Our findings support the notion that board network has a higher marginal effect on financial stability for banks which faces more severe asset quality problems.

This study contributes to the literature in several ways. First, our study contributes to the social network literature. There is an extensive body of research on the understanding of organizational networks (Borgatti and Cross, 2003). A significant proportion of these studies focus on non-financial and its relation to organizational outcomes. Larcker et al. (2013) find that the more central firms in the network enjoy superior performance. Director networks also affect the acquisition behavior of the firms (Renneboog and Zhao, 2014), governance practices (Fracassi and Tate, 2012), and innovation (Helmets et al., 2017) and others. In the context of financial firms, Hochberg et al. (2007) find that more connected venture capital firms experience better performance given by a higher likelihood of exiting the investment through a successful IPO route. Few studies (Houston et al. 2018; Abdelbadie and Salama, 2019) have focused on bank networks in the context of developed countries like the United States, Europe, and Australia. Most of the developed economies have a market-based financial system with developed capital markets, which is an additional information channel available to the banks for assessing potential borrowers. However, in emerging market economies with less than fully developed capital markets and bank-based financial system, the role of bank board networks becomes even more important as an information channel. Our study focuses on the bank-board networks in the context of India; an emerging market economy provides further evidence that formal social networks due to board interlocks matter for transmission of soft information among firms.

Second, we contribute to corporate governance in banks literature by examining how board network affect financial stability of banks. Our study is closely related to the studies by Houston et al. (2018) and Abdalbadie and Salama (2019). The former study shows that more central banks in the global banking industry are likely to play a lead role in a syndicated loan origination process. The latter study highlights the relation between bank board network and risk-taking behavior of banks in the United States. This study only considers the connection among banks due to the indirect connections of the bank director. On the other hand, our study

considers all direct and indirect connections of the bank directors. This is important to understand the diversity of information available to the bank through the board network.

Finally, our study provides new evidence regarding the importance of board network for banks in India. Previous studies have highlighted the importance of board networks for non-financial firms in India (Biswas, 2019; Shaw et al., 2016). This study empirically shows that well-connected directors sitting on bank boards can provide valuable strategic advice and help in improving the asset quality of banks. This finding has policy implications regarding the board composition of banks. The central bank in India – Reserve Bank of India – in 2014 Review of Governance of Boards of Banks in India Report emphasized the importance of independence and skillset of the director for improving board quality, but what constitutes skillset is not clear. Our results suggest that the connections of the director could improve the ability to provide strategic advice and should be considered as a desirable board attribute of the bank director.

The remaining of the paper is organized as follows. Section 2 develops the conceptual framework of the study. Section 3 discusses the data and variables. Section 4 presents the methodology employed in the study. Section 5 presents the financial stability results. Section 6 discusses the implications of the study and concludes.

2. Conceptual framework

The resource dependence theory (Pfeffer, 1972) and the agency theory (Jensen and Meckling, 1976) are the two prominent theories explaining the role of the board in firms. According to the resource dependence view, the board of directors helps organizations to manage external uncertainty and provide access to external information and other resources. On the other hand, the agency theory of board emphasizes the disciplinary role of the board wherein the board act as monitors on behalf of the shareholders. The agency and resource dependence theories are often viewed as competing theories of the board as there is a trade-off between the role of the board as monitors and resource providers. However, Hillman and Dalziel (2003) argue that there is a need to integrate the resource dependence and agency views to understand the role of the board in modern corporations. This view suggests that firms with higher board capital are likely to perform better as they have access to a large pool of resources. Also, the more experienced and connected directors are better able to monitor the managers, given their vast experience and skillset.

2.1 Board network and firm outcomes

One aspect of the board that has received considerable attention in the literature is the importance of board network on account of director interlocks (see Lamb and Roundy, 2016 for review). Firms having a well-connected board are found to be more profitable (Larcker et al., 2013; Hochberg et al., 2007). The connected directors help in the transmission of soft information and give their superior skillset are better strategic advisers to the firm. The reputational hypothesis suggests that the busy director occupies multiple board seats due to their ability and have an established reputation. Such connected directors are likely to be better monitors and improve board effectiveness. The literature on busy boards is closely related to the board network literature and highlights the dark side of board networks. The interlocked director occupying multiple board seats is considered busy having less time to devote to each firm, and this negatively affects the performance of the firm (Li and Ang, 2000). Barnea and Guedj (2009) find that interlocked firms pay higher CEO salaries, CEO pay is less sensitive to performance, and CEO turnover is less likely.

Few studies have also examined the role of social networks in the context of banks. The bank CEOs with large social network increases bank risk-taking behavior (Dbouk et al., 2020) finding evidence in favor of weak monitoring hypothesis. The connections of the CEOs reduce the search cost associated with job search in case the CEO is unable to perform in their current role. This implicit insurance increases the risk-taking behavior of the CEOs with a large social network. On the other hand, Abdelbadie and Salama (2019) find that more connected bank boards are likely to reduce the risk-taking behavior of banks in the United States. This study finds evidence in favor of the integrated resource dependence hypothesis. A large board network improves the quality of advice provided by the director to the bank, and the connected directors are better able to monitor the banks.

Figure 1 below presents a conceptual framework depicting the relation between bank board network and the financial stability of banks. The first channel suggests that board connectedness could adversely affect the financial stability of banks. The connectedness of the board is likely to be high when several directors occupy multiple board seats. The busyness hypothesis suggests that the connected director will be busy and will be able to devote less time to board activities. Given the complexity of banking operations, lack of time devoted by the director can have a negative effect on board oversight function of the board and the financial stability of banks. Further, the well-connected director will know CEOs and other board

members of various firms, which secures the career progression of the connected director. The second channel suggests that the well-connected bank director is a proxy for the ability of the director. The ability of the director will overcompensate for the lack of time, as suggested by the busyness hypothesis. The monitoring ability of bank director is crucial for the board effectiveness of banks. The third channel suggests that more connected bank-director will have access to a large pool of resources. The access to soft information may reduce the information asymmetry between the bank and its borrowers, especially in economies with less than perfect capital markets. The final channel suggests that the higher expertise of the connected director makes them better strategic advisors, improves board effectiveness and financial stability of banks.

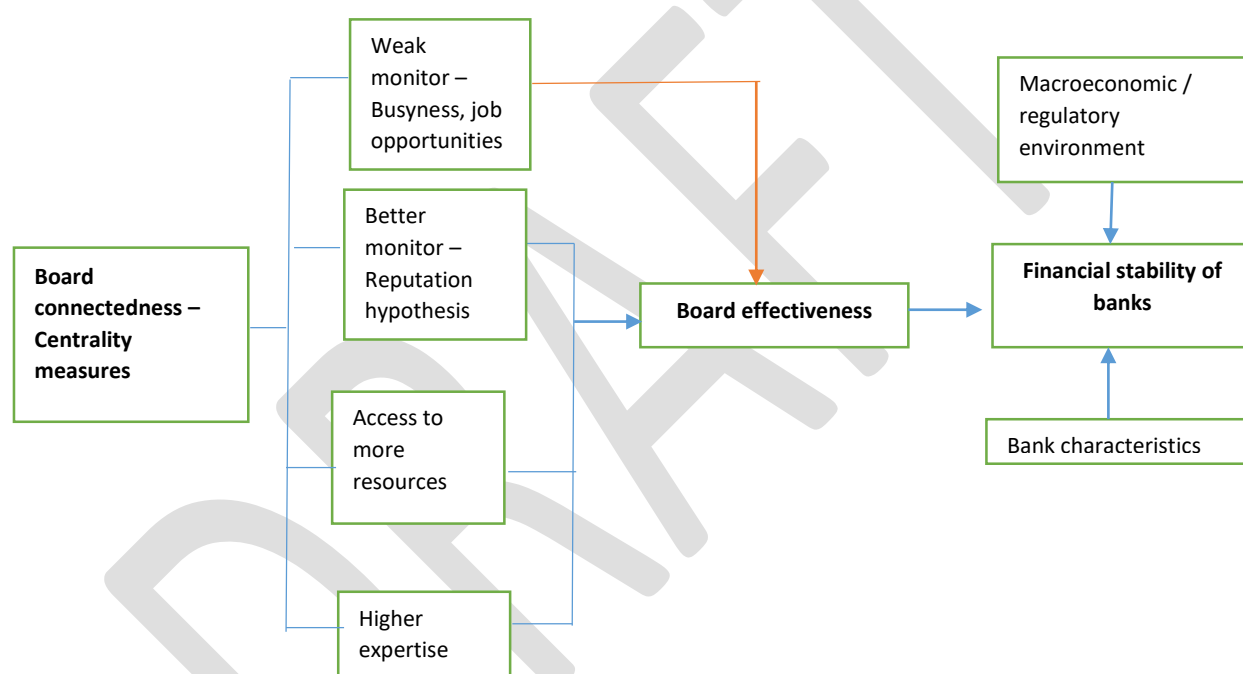


Figure 1: Conceptual framework

2.2 The banking sector in India

The Indian banking sector is dominated by 25 large public sector banks (PSBs) where the government is the majority shareholder and 21 private banks. The foreign banks in India operate as branches. The asset quality review initiated by the Indian central bank, the Reserve Bank of India, brought the burgeoning non-performing assets (NPAs) problem in the limelight. The NPAs were significantly higher for the PSBs than the private banks in India. The deteriorating asset quality of Indian banks came re-ignited the need to focus on good governance and role on bank boards in monitoring the risk-management practices and financial stability of banks.

The bank board comes under the regulatory supervision of the Banking Regulation Act, 1949 (Reserve Bank of India), the Companies Act, 2013 and the Security and Exchange Board of India (SEBI). The Banking Regulation Act is specially designed for all banks operating in India, and eliminating conflict of interest is at the heart of its provisions. The Companies Act, 2013 is applicable for all companies operating in India, and its provisions are meant for the smooth functioning of a company. SEBI Act, 1992 was enacted to provide the establishment of a board to protect the interests of investors in securities and to promote the development of and to regulate the securities market and matters connected therewith. In all the above mentioned Acts, there are clauses/ sections which deal with corporate governance of an entity/ bank in general and board structure in particular.

Section 10(A) of Banking Regulation Act, 1949, Section 149 of the Companies Act, 2013 and Clause 49 Listing Agreement (SEBI) deals with the Board of Directors of an entity. With respect of banks, Banking Regulation Act, 1949, supersedes all other regulatory requirements. As per Section 10(A) of Banking Regulation Act, 1949, not less than 51% of the directors need to have special knowledge or practical experience in areas such as accountancy, agriculture and rural economy, banking, cooperation, economics, finance, law, and small-scale industry. This section also makes provisions concerning composition as well as the tenure of a director on the bank board, including the chairman. Similarly, Section 149 to 152 of the Companies Act, 2013 and Clause 49 of Listing Agreement deal with composition of board and different committees. Further, the PSBs are also regulated by the Ministry of Finance, Government of India. The Nayak Committee report (2014) on the Review Governance of Boards of Banks in India highlighted that the bank boards of PSBs are disempowered and are characterized by weak corporate governance structure. In recent years, the emphasis has been put on the significance of independent directors on the board; however, all these regulations does not discuss the significance of bank board network in corporate governance.

2.3 Research question

Given the backdrop of deteriorating asset quality of Indian banks and corporate governance challenges faced by the banks, we consider the role of the director network on the financial stability of banks in India. The conceptual model discussed earlier suggests that that the bank board network on account of director interlocks will enhance the monitoring ability of the board, which in turn will improve the financial stability of banks. Further, a large number of board seats occupied by the bank-director makes them better advisors and may improve the

lender-borrower matching for the bank. Both these channels suggest that bank-board with a large number of direct connections should be related to the improved financial stability of banks. On the other hand, the connected directors will increase the resource pool of the bank, both through direct and indirect connections. A large indirect board network will imply access to soft information regarding various industries and potential borrowers. The resource channel suggests that the direct and indirect connections of the bank board should be related to the higher financial stability of banks.

3. Data and Variables

3.1 Data

The direct and indirect connections are constructed using the data obtained from the Indian Boards database maintained by NSE Infobase. This database provides information regarding the names of the board members, appointment and cessation dates, other directorships positions (both listed and unlisted), age, type of director, education, and gender for list of companies/banks from 2014. Since the data is not available for the board composition of unlisted banks, we only consider the listed commercial banks for the analysis.

The financial data for the banks are obtained from the Prowess database maintained by the Centre for Monitoring Indian Economy (CMIE). Our sample consists of 43 commercial banks in India for the period 2009-2019.

3.2 Board network variable

Using the board level data, we obtain direct and indirect connections at the bank-director level and then construct the measures at the bank level. The first level degree measure gives direct connections. This is a count measure and gives the number of directors with whom each of the bank director is connected. The first level degree measure for bank i at the end of year t is given by:

$$DEGREE_1_{it} = \frac{\sum_{j=1}^k Connections_{jt}}{Board\ size_{it}}$$

where $Connections_{jt}$ is the total number of director level connections of j^{th} bank director during year t . $Board\ size_{it}$ is the board size of bank i in year t .

The indirect connections of the bank due to director interlocks are given by degree, closeness, betweenness centrality measures. Each of the centrality measure captures a different aspect of the network (Omer et al., 2014). The second level degree measure is the number of

direct connections of the bank director's first level connections. DEGREE_2 variable is defined as the ratio of the sum of the second connections level of all the directors on a bank board to the board size. The first and the second level degree measures are proxies of the quantity of information available to the bank through the bank directors.

The closeness measure (CLOSE) is given by the inverse of the sum of the distance between a bank director from all other directors in the network. The banks with a higher value of closeness measure are likely to obtain soft information faster than the banks with a lower value of closeness measure. If for a bank, all the directors are only sitting on the board of a particular bank and do not occupy any other board seat, then the closeness measure is defined as zero. The closeness measure (CLOSE) is the sum of the closeness of the directors of a bank divided by the size of the board.

$$Closeness_director_{jt} = \frac{1}{\sum_{\substack{s=1 \\ s \neq j}}^k Distance_{js}}$$

$$CLOSE_{it} = \frac{\sum_{j=1}^k Close_director_{jt}}{Board\ size_{it}}$$

The betweenness measure (BTWN) is defined as the sum of the number of paths that pass through a bank-director for connecting any two directors in the network. This measure captures the importance of an agent in the network. When a large number of paths crosses through the bank director, the bank director becomes crucial for information transmission within the network. Similarly, a bank with a high betweenness score will have an informational advantage in the network over the other banks.

The eigen measure captures the power and prestige of the bank director in the network. The eigen measure captures to what extent the bank director is connected to other well-connected directors. The eigen centrality measure (EIGEN) is defined as the sum of eigen centrality values of the directors of the bank divided by board size.

Since, each centrality measure captures a different aspect of the network, in the regression analysis we compute the z-score of the bank-level centrality measures for interpreting the coefficients. Further, the centrality measures are likely to be correlated and are included in separate models.

3.3 Nature of bank board network variable

Based on the networking framework discussed in the previous section, we can determine different aspects of the board network at the director-level and also at the bank-level. The board data provides an alpha-numeric character that is uniquely associated with a director and remains constant over time. The selection of this particular is based on the fact that this director has a minimum number of directorship (greater than 1) in a bank board. In terms of examining the network of a director, we perform the analysis at two levels. The first-level of connection of a director is established when the bank director occupies a board seat. The second-level connection of the bank director occurs when the given director is connected to other directors of different companies via their first-level connections.

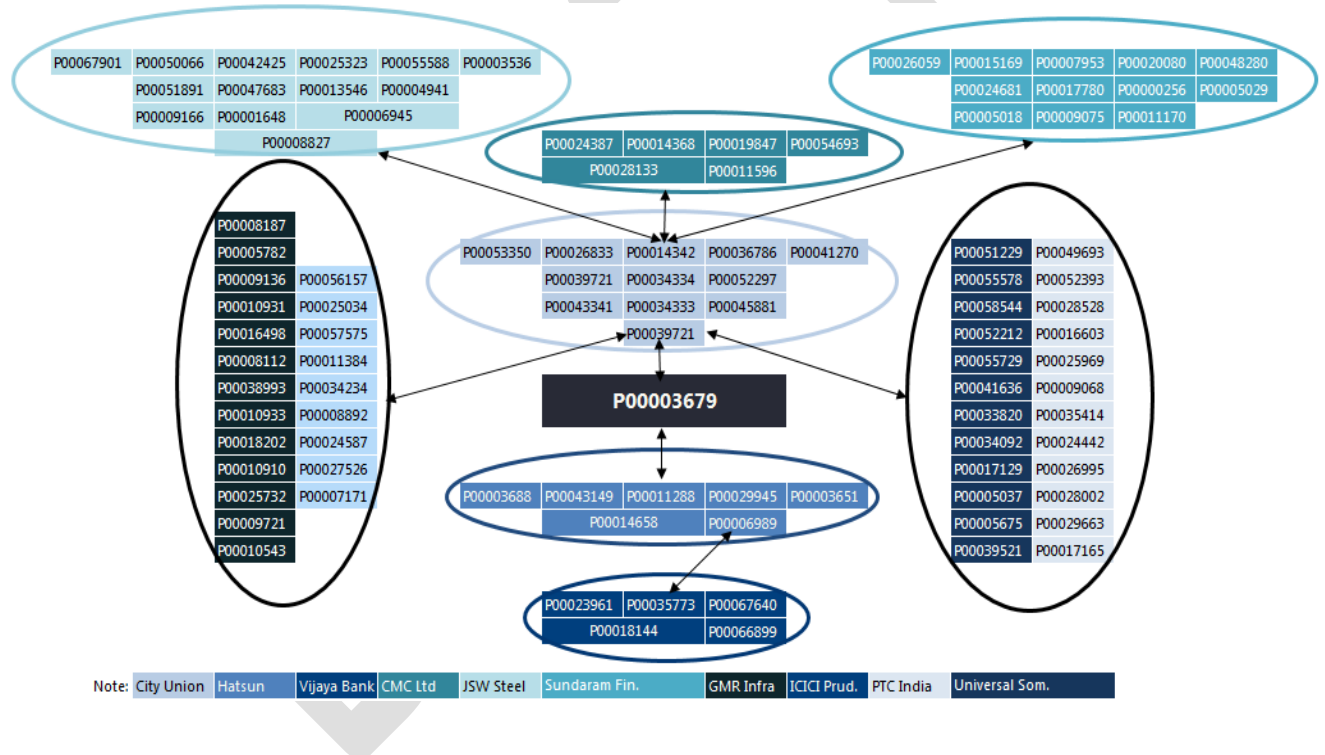


Figure 2: Director-level network

Figure 2 above shows the network of director P00003679, who is sitting on the board of City Union Bank Ltd. (City Union) and Hatsun Agro Product Ltd (Hatsun). At the first-level connection, the given director is connected to all the directors sitting on the board of City Union and Hatsun. Therefore, the given director is connected to 19 different directors (12 directors of City Union and 7 directors of Hatsun) which constitutes the direct connections. Now two directors of City Union (P00014342 and P00039721) and one director of Hatsun (P000069689)

are sitting on board of different companies. ‘P00014342’ of City Union is sitting on the board of CMC Ltd, JSW Steel, and Sundaram Finance, whereas ‘P00039721’ of City Union bank is sitting on the board GMR Infra., ICICI Prudential and PTC India. Similarly, ‘P00006989’ of Hatsun is sitting on the board of Universal Som. The given director is connected to 32 directors via ‘P00014342’ and 46 directors via ‘P00039721’. The given bank director is connected to 5 directors via ‘P00006989’. Therefore, at the second-level, the bank director is connected to 83 directors in total, 78 different directors through its board membership at City Union Bank Ltd. and 5 different directors due to its directorship at Hatsun Agro Product Ltd. Therefore, by combining both first and second level networking, the given director is connected to 102 different directors. Figure 2 shows that the given bank director is connected to 102 different directors, sitting across eight different companies varying from financial to infrastructure to steel industries. The director-level board network measure will provide both the quantity of information available to the bank and the strength of the network. The strength of the network can increase the quality of information that is transmitted to the bank. Our board network measures in the main analysis capture the joint effect of the importance of information quantity and quality of information on financial stability of banks.

Due to limitations of the data, we cannot go beyond the second level connection, however, they exist in banking ecosystem and hence it is imperative that regulators take into their consideration while formulating regulations related number of directorship position by a director.

We also consider the network at the bank-level to check the consistency of our findings obtained using the director-level board network measures. Figure 3 presents the network of City Union Bank Ltd when the network is computed directly at the bank-level and not at the director-level.

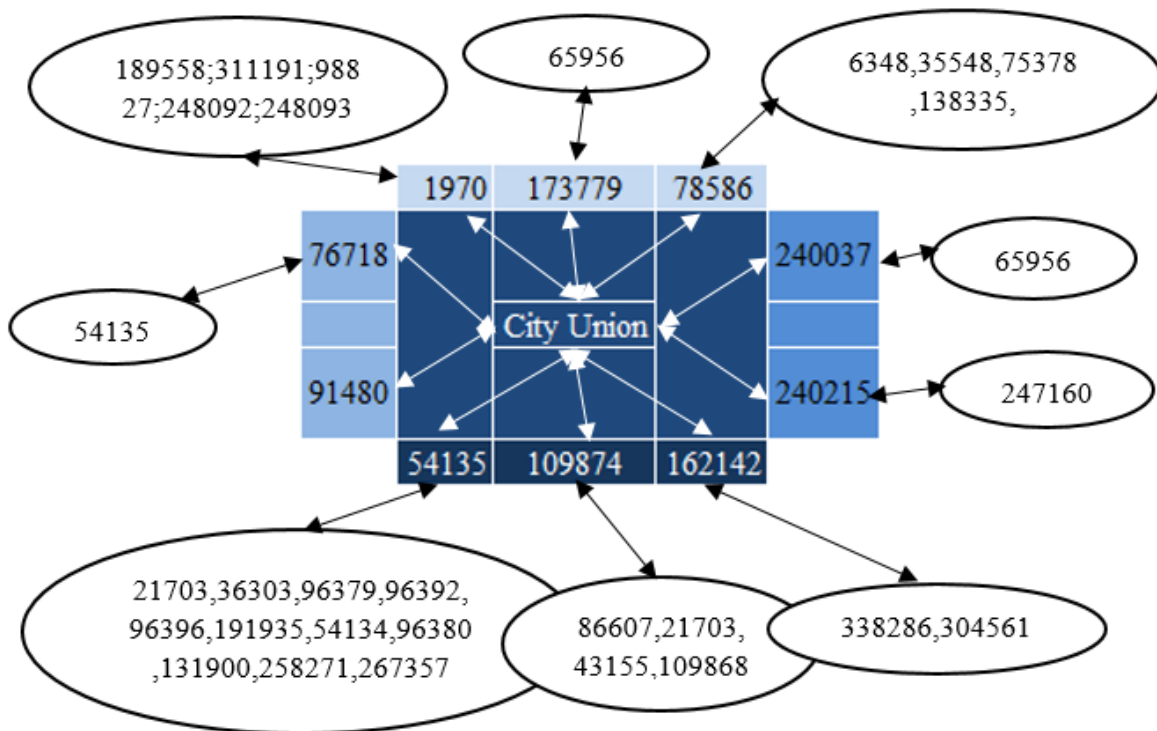


Figure 3: Board network at the bank-level

The network at the bank-level provides a different perspective on bank board interlocks. Similar to director interlocks of City Union, the bank is directly connected with ten different companies (direct connections) and is connected to 32 other companies at the second-level (indirect connections). This measure provides a unique number of companies with which the bank is directly or indirectly connected due to board interlocks. This measure captures the pool of diverse resources available to the bank.

3.4 Other variables

Dependent variable: The dependent variable in the analysis is the financial stability of the bank given three different measures. The first measure is an asset quality measure that captures the gross and net non-performing assets of the bank. A credit extended by the bank is classified as a non-performing if the borrower has not paid interest or principal for over 90 days. A higher level of non-performing assets indicates that the bank is under stress as the likelihood borrower default has increased. This measure is also regularly tracked and reported by the RBI to comment regarding the financial stability and resilience of the banks in the country. We consider the ratio of gross non-performing assets to total advances (GNPA) and the proportion of net non-performing assets to total advances (NNPA) as the dependent variables in the study. These measures of credit risk outcomes have been considered in other bank studies in the

literature (Abdelbadie and Salama, 2019; Sarkar and Sarkar, 2018; O’Sullivan et al., 2016). The second measure is the z-score defined as the sum of return on assets (ROA) and capital to assets ratio (CAR) divided by the standard deviation of ROA (σROA). This measure gives the distance from default, and higher value suggests that the bank has sufficient profit to cover its debt obligations, hence, less likely to default. This measure of financial stability has been used by several studies (Abdalbadie and Salama, 2019; Ahamed and Mallick, 2017; Elyasiani and Zhang, 2015; Laeven and Levine, 2009). The z-score variable is highly skewed, and we consider the logarithm of the z-score for the analysis. Finally, the σROA measure gives the variation in bank’s ROA for the last 5-year period. This measure provides the volatility of the accounting performance of the bank, and a lower value suggests a more stable financial outlook (Ahamed and Mallick, 2017; Laeven and Levine, 2009).

Control variables: We control for a set of bank characteristics in line with existing literature (Abdalbadie and Salama, 2019; Louzis et al, 2012). The bank size (SIZE) given by the logarithm of total assets controls for diversification and economies of scale. We also control for the debt-equity ratio of the bank (LEVERAGE), bank funding source given by logarithm of deposits (DEP), the efficiency of banks captured by profit per employee (EFFICIENCY), and the net interest margin (NIM).

4. Methodology

We estimate a panel data model to analyze the relation between board network variable and financial stability measures given by the following specification:

$$Stability_{it} = \beta_0 + \beta_1 Board\ Network_{it-1} + \beta_2 Ownership_i + \sum_{j=1}^k \gamma_j X_{ijt-1} + \varphi_t + \varepsilon_{it} \quad (1)$$

where $Stability_{it}$ is the asset quality measure (GNPA1, GNPA2, NNPA1 and NNPA2), logarithm of z-score and σROA variables. The $Board\ Network_{it-1}$ is the lagged value of the z-score of the direct (z-DEGREE_1) and indirect connections of the bank (z-DEGREE_2, z-CLOSE, z-BETWEEN, z-EIGEN), $Ownership_i$ is the dummy which takes the value one for private banks and zero for public sector banks, X_{ijt-1} is the matrix of lagged control variables, φ_t is the year fixed effect and ε_{it} is the random error term.

The empirical results obtained from estimating the above specification will be biased and inconsistent if the board network measures are endogenous. The endogeneity could be due to omitted variables, reverse causality, or simultaneity in the system. One can argue that the more stable banks can attract well-connected directors on their board, leading to reverse

causality. Further, the reputation of the bank in the marketplace can be related to both the financial stability of the bank and the number of directorships of the bank director. We address the endogeneity concerns using two approaches. First, we lag the board network measures by one-year, so that the network values are pre-determined. Second, we also employ an instrumental variable (IV) approach to ensure that the coefficients are consistent. The two-step least squares (2SLS) method in the presence of a valid instrument can tackle the endogeneity in the model. A valid instrument should be correlated with the board network variable and should be otherwise uncorrelated to the financial stability measures.

We use board size as an instrument for the board network variable (Abdalbadie and Salama, 2019). Theoretically, larger boards increasing the chances of bank director occupying multiple board seats and this should be related to the number of direct and indirect connections of the bank. We expect a positive and significant relation between the board network measures and the board size variable. However, board size should not be directly related to the financial stability of the banks in the sample. Prior studies have also not found any conclusive relation between board size and bank outcomes (Sarkar and Sarkar, 2018; Talavera et al., 2018; Akbar et al., 2017).

In the first stage of 2SLS estimation, we regress the board network measure on the logarithm of board size variable and the other bank controls along with year fixed effects.

First stage equation:

$$Board\ Network_{it} = \alpha_0 + \alpha_1 Board\ size_{it} + \alpha_3 Ownership_i + \sum_{j=1}^k \phi_j X_{ijt} + \vartheta_t + u_{it} \quad (2)$$

In the second stage, the financial stability measure is regressed on the predicted board network variables obtained from the first stage and other controls.

Second stage model:

$$Stability_{it} = \beta_0 + \beta_1 \widehat{Board\ Network}_{it-1} + \beta_2 Ownership_i + \sum_{j=1}^k \gamma_j X_{ijt-1} + \varphi_t + \varepsilon_{it} \quad (3)$$

5. Results

5.1 Summary statistics

Figure 4 presents the overtime trend in average direct and indirect connections of the banks on account of board interlocks. The direct links of banks have remained stable during the last decade; however, the second-level connections of banks have increased over-time. In 2009, the average number of director connections of the bank at the second-level was 127, which

increased to around 173 second-level connections in 2019. The increasing nature of indirect connections reinforces the need to take into account indirect connections when analyzing the importance of board network on outcomes.

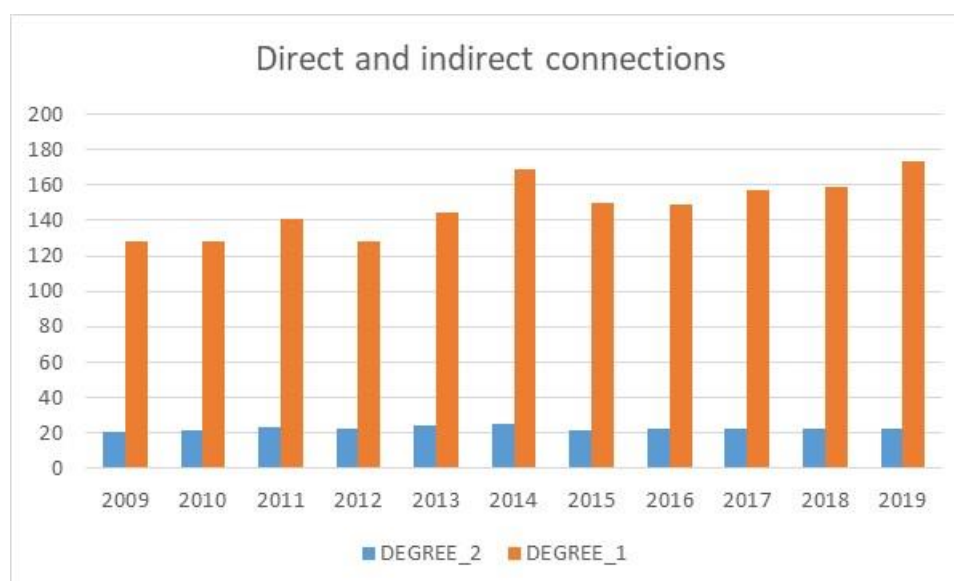


Figure 4: Director level connections of bank boards since 2009

Table 1 presents the summary of the financial variables for all the banks during the period and also separately for the PSBs and the private banks. It is evident that during the study period, the non-performing assets in the Indian banking industry was high and the average non-performing assets were much higher for the PSBs compared to the private sector banks in India. The PSBs in India are less connected than the private banks in terms of both direct and indirect connections of the board. The average size and deposits of the PSBs are higher than the private counterpart. However, the ROA, NIM, and EFFICIENCY of the private banks are higher than the PSBs.

<<Insert Table 1 here>>

5.2 Director-level board network and asset quality

Table 2 presents the 2SLS results for the regression of gross non-performing assets to total advances (GNPA) on the network measures. The first stage regressions (Columns 1,3,5,7, and 9) ascertains that the board size is a relevant IV and is positively related to the board network measure. The F-statistic of the first stage regressions are greater than ten, indicating that the instrument is not weak. The p-value of the Wu-Hausman test suggests that the pooled estimates would be inconsistent due to the endogeneity of board network measures. The second stage

regressions indicate that board network is related to the asset quality of the bank. For example, column 2 suggests that the one standard deviation increase in z-DEGREE_1 measure is related to 1.3 percentage points fall in GNPA. The indirect board network measures (z-DEGREE_2, z-EIGEN, z-CLOSE, z-BTWN) are important determinants of the bank's asset quality, even though the marginal effect is highest for the z-EIGEN measure. We do not find that a particular aspect of the network like the speed of information or the quantity of information available to the bank, dominates the results.

<<Insert Table 2 here>>

In a bank-based economy, the human capital and resources of the bank-directors could be crucial for the quality of strategic advice provided to the bank. The well-connected director will be better informed regarding the borrower's (mainly corporate borrowers) credit behavior and help in reducing the information asymmetry between the bank and its borrowers. Further, knowledge regarding potential borrowers through connections can also lower the probability of extending loans to risky borrowers, which in turn is related to lower non-performing assets.

Table 3 presents the results for the net non-performing assets measure (NNPA). Similar to the results for the gross non-performing asset measures, we find that a larger board network is beneficial for the bank's asset quality. One standard deviation increase in z-DEGREE_1 is related to 0.5 percentage points fall in NNPA (column 1). An increase in indirect connection measures are also related to 0.5-0.6 percentage points fall in NNPA (columns 2-5). Additionally, we find that the average non-performing asset is lower for private banks, and banks with higher deposits, higher net interest margin and lower leverage. Further, we find that bigger banks have higher non-performing assets.

<<Insert Table 3 here>>

The results in this section indicate that the access to valuable resources and information available through the board network and the expertise of the well-connected director can help in improving the asset quality of banks in India. The findings are in line with the integrated resource dependence view of boards. These results are in line with other bank-specific studies in the context of the United States (Adelbadie and Salama, 2019) wherein a higher board network is related to the lower credit risk of banks. These results also confirm the findings of other studies where busy bank directors are related to lower risk-taking by banks (Elyasiani and Zhang, 2015), refuting the busyness hypothesis for banks in the United States.

5.3 Director-level board network and z-score

Next, we consider the logarithm of z-score as a measure of financial stability of banks and the results are presented in Table 4. We find that one standard deviation increase in z_DEGREE_1 is related to 19.3% fall in insolvency risk (column 1), and one standard deviation increase in indirect connections is related to 19.1% to 24.9% fall in the risk of insolvency (columns 2-5). The findings indicate that the well-connected bank-directors bring in financial expertise and can help the banks to manage their risk better and lower the probability of insolvency. These findings are also in line with the study by Carpenter and Westphal (2001), which highlights that ties of the board improve the board's ability to supervise and advise the management effectively. This effective functioning of bank boards can reduce the insolvency risk of the bank. Abdalbadie and Salama (2019) also find that an increase in the board network is related to a fall in insolvency risk of the banks, and the effect size was in their study was in range of 47%-59%.

<<Insert Table 4 here>>

5.4 Director-level board network and volatility of ROA

The third financial stability measure in our study is the volatility of ROA during the last five years. The results of using the accounting measure of risk as the dependent variable are presented in Table 5. We find that one standard deviation increase in z_DEGREE_1 is related to 6.1 percentage points fall in σROA (column 1). One standard deviation increase in the z-score of indirect centrality measures is related to 6.1 to 7.9 percentage points fall in σROA (columns 2-5). The effect of an increase in indirect connections is at least as large as the effect of an increase in direct connections. The advice of the well-connected bank director can help the banks to smoothen their performance during peaks and troughs of a business cycle. Also, the availability of soft information on account of a larger board network can help banks perform better even when the macroeconomic environment is not favorable.

<<Insert Table 5 here>>

5.5 Public and private banks

Our main results suggest that a well-connected bank board improves the financial stability of banks in India. However, the regression coefficients give us an average effect of board network for the listed banks in India, and the effect could be different for sub-groups based on ownership. The Nayak Committee Report (2014) emphasized regarding the deteriorating asset

quality, especially in PSBs in India. During our study period, the average non-performing PSBs were higher than that of private banks for all the four asset quality measures (Table 1). If a larger board network helps the bank by providing crucial information and a well-connected director is better able to monitor the bank activities, then the marginal effect of board network should be higher for PSBs having inferior asset quality than for private banks. Table 6 presents the results obtained from separately estimating the relation between NNPA and board network measures for PSBs and private banks. We find that an increase in board network is related to a fall in NNPA for the PSBs only, and there is no significant effect for the sample of private banks³. The full sample results seem to be driven by the set of PSBs in India. These results largely confirm the findings of Larcker et al. (2013) they find that board network is more beneficial for the less profitable firms in the United States.

<<Insert Table 6 here>>

5.6 Alternate network measure

In our analysis, we have computed the board network by standardizing the sum of the network of the directors sitting on the bank board. Now, we consider an alternate definition of the board network variable by calculating the network measures at the bank level on account of director interlocks. The discussions in section 3.3 earlier highlight that the director-level board network measures capture the quantity and strength of bank connections. On the other hand, the bank-level board network provides a measure of the availability of a unique set of information to the bank.

We re-examine the relation between board network and financial stability of the bank by computing the board network at the bank level instead of the director level. Table 7 presents the 2SLS results obtained from regressing GNPA variable on bank-level board network measures and other controls. The coefficient of *Board size* variable is positive and significant at 1% level of significance, indicating a strong positive relation between bank-level board network and the instrument. Further, the first stage F-statistic values are much higher than the threshold level of 10, suggesting that the instrument is not weak-instrument. The bank-level board network measures have an effect of lowering the GNPA at the bank level on an average⁴.

³ For brevity we report the results for the NNPA alone. For z-score and σROA measures the board network variable is significant only for the public sector banks sample, however, for the GNPA measure the board network is negatively related for both public sector and private banks. The tables are available upon request.

⁴ For brevity we report the results for GNPA alone, however, the results are similar for NNPA, z-score and σROA and are available upon request.

Our main results in the previous section are robust to this alternate definition of the board network variables.

<<Insert Table 7 here>>

6. Discussions and conclusions

Since the GFC, the policymakers have laid great importance to the corporate governance of banks to ensure that banks do not engage in excess risk-taking behavior. Several studies have examined the link between various aspects of the governance structure of banks, including ownership, board structure, and incentives of the CEOs. In this study, we examine the role of the board network of banks and its relation with financial stability in the context of an emerging market economy like India. We consider the listed banks in India and construct the first and second-level connections between bank directors and other directors sitting on listed as well as unlisted companies with the help of appointment and cessation dates for the period 2009-2019. We find that the indirect connections of the banks have increased by around 40% during the last decade. Further, the public sector banks in India are less connected than private banks.

The direct and indirect connectedness of banks is related to its financial stability. We find that a well-connected bank board helps banks to improve asset quality, reduce the insolvency risk, and also reduce the volatility of ROA. The findings of the study provide evidence in support of the integrated resource-based view of the board. The well-connected bank director provides access to soft information, which helps to reduce the information asymmetry between the bank and potential borrowers. The experienced connected director may provide strategic advice and help the bank to reduce the volatility in performance. Finally, well-connected directors are likely to be effective monitors and lower the insolvency risk of the bank. Further, we find that the board network is more important for public sector banks facing serious asset quality issues in the past.

The findings of this study have implications for regulators and investors. The Companies Act, 2013, capped the maximum number of board seats that can be occupied by the directors of listed firms in India at 15. Our study shows restrictions for bank directors may not yield net benefit. We find that the benefit of the board network of the bank on account of multiple directorships can outweigh the costs of interlocks, especially for public sector banks in India. Further, our findings suggest that leveraging the social capital of the board could one of the various means that banks can adopt to address the growing non-performing asset

problem. For investors, our study indicates that banks having a large board network are likely to have low insolvency risk, and this should be related to a lower unsystematic risk of the stock.

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Table 1: Summary statistics

	All Banks		Public Sector Banks		Private Banks	
	Mean	Standard Deviation	Mean	Standard Deviation	Mean	Standard Deviation
GNPA	6.656	6.132	9.745	6.566	2.870	2.213
NNPA	2.685	2.965	3.737	3.433	1.315	1.265
ROA	0.653	1.006	0.320	0.968	1.096	0.878
NIM	3.038	1.082	2.660	0.556	3.616	1.387
z- DEGRE_1			-0.265	0.659	0.319	1.241
z- DEGREE_2			-0.360	0.627	0.430	1.193
z-EIGEN			-0.346	0.476	0.431	1.287
z-CLOSE			-0.321	0.714	0.375	1.168
z- BETWEEN			-0.324	0.658	0.377	1.209
Leverage	1.316	0.923	1.330	0.785	1.299	1.075
Size	13.890	1.549	14.526	0.846	13.091	1.839
Efficiency	1.100	6.395	1.070	6.814	1.140	5.807
Deposits	14.267	1.422	14.901	0.832	13.460	1.599

Table2: Board network and GNPA of banks

	<i>Dependent variable: GNPA</i>									
	1 st stage (1)	2 nd Stage (2)	1 st stage (3)	2 nd Stage (4)	1 st stage (5)	2 nd Stage (6)	1 st stage (7)	2 nd Stage (8)	1 st stage (9)	2 nd Stage (10)
Board size	1.516*** (0.005)		1.438*** (0.013)		1.174*** (0.032)		1.529*** (0.006)		1.385*** (0.002)	
z-DEGREE_1		-0.013*** (0.004)								
z-DEGREE_2				-0.013*** (0.004)						
z-EIGEN						-0.018*** (0.006)				
z-CLOSE								-0.013*** (0.004)		
z-BTWN										-0.014*** (0.004)
PRIVATE	1.001*** (-0.035)	-0.052*** (0.008)	1.131*** (-0.045)	-0.050*** (0.008)	1.217*** (-0.102)	-0.041*** (0.011)	0.971*** (-0.047)	-0.054*** (0.007)	0.992*** (-0.045)	-0.053*** (0.008)
LEVERAGE	0.114*** (-0.009)	-0.019*** (0.005)	0.002 (-0.008)	-0.021*** (0.005)	-0.038*** (-0.006)	-0.022*** (0.005)	-0.007 (-0.013)	-0.021*** (0.005)	0.068*** (-0.013)	-0.019*** (0.005)
NIM	0.036*** (-0.008)	-0.013*** (0.003)	0.030*** (-0.009)	-0.013*** (0.003)	0.002 (-0.019)	-0.013*** (0.003)	0.043*** (-0.010)	-0.012*** (0.003)	-0.018*** (-0.008)	-0.013*** (0.003)
SIZE	-0.480** (0.269)	0.135*** (0.044)	0.684** (0.306)	0.150*** (0.045)	0.372 (0.365)	0.149*** (0.046)	0.780** (0.343)	0.152*** (0.044)	0.185 (0.342)	0.146*** (0.045)
EFFICIENCY	-0.007***	-0.001**	-0.002***	-0.001*	-0.0001	-0.001*	-0.002***	-0.001**	-0.003***	-0.001**

	(0.0001)	(0.0003)	(0.0002)	(0.0004)	(-0.0002)	(0.0004)	(0.0001)	(0.0004)	(0.0001)	(0.0004)
DEPOSITS	0.549*	-0.129***	-0.572*	-0.144***	-0.258	-0.140***	-0.708*	-0.147***	-0.128	-0.141***
	(-0.299)	(0.045)	(-0.344)	(0.045)	(-0.453)	(0.047)	(-0.383)	(0.045)	(-0.379)	(0.046)
Constant	-5.912***	0.083	-5.875***	0.081	-5.144***	0.046	-5.506***	0.093*	-5.061***	0.089
	(0.574)	(0.053)	(0.661)	(0.055)	(1.397)	(0.067)	(0.752)	(0.052)	(0.699)	(0.054)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	342	205	342	205	342	205	342	205	342	205
Adjusted R ²	0.613	0.692	0.622	0.680	0.461	0.639	0.602	0.686	0.532	0.664
First stage F-stat		253.429		192.130		65.580		211.080		146.620
Wu-Hausman test (p-val)		0.000		0.000		0.000		0.001		0.000

* p<0.1; ** p<0.05; *** p<0.01

Table 3: Board network and NNPA of banks

	<i>Dependent variable: NNPA</i>				
	(1)	(2)	(3)	(4)	(5)
z-DEGREE_1	-0.005*** (0.002)				
z-DEGREE_2		-0.005*** (0.002)			
z-EIGEN			-0.006*** (0.002)		
z-CLOSE				-0.005*** (0.002)	
z-BTWN					-0.005*** (0.002)
PRIVATE	-0.022*** (0.003)	-0.021*** (0.003)	-0.019*** (0.004)	-0.022*** (0.003)	-0.022*** (0.003)
LEVERAGE	-0.008*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)	-0.009*** (0.002)
NIM	-0.007*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)	-0.007*** (0.001)	-0.008*** (0.001)
SIZE	0.057** (0.023)	0.063*** (0.024)	0.062** (0.024)	0.063*** (0.024)	0.061** (0.024)
EFFICIENCY	-0.0001* (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001* (0.0001)
DEPOSITS	-0.058** (0.024)	-0.064*** (0.024)	-0.062** (0.025)	-0.064*** (0.024)	-0.061** (0.024)
Constant	0.081*** (0.024)	0.081*** (0.024)	0.079*** (0.026)	0.084*** (0.024)	0.084*** (0.024)
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	340	340	340	340	340
Adjusted R ²	0.638	0.634	0.614	0.633	0.620
First stage F-stat	307.200	260.212	113.297	278.079	207.56
Wu-Hausman test (p-val)	0.007	0.004	0.002	0.003	0.000

* p<0.1; ** p<0.05; *** p<0.01

Table 4: Board network and insolvency risk

	<i>Dependent variable: logarithm z-score</i>				
	(1)	(2)	(3)	(4)	(5)
z-DEGREE_1	0.193** (0.084)				
z-DEGREE_2		0.203** (0.090)			
z-EIGEN			0.249** (0.112)		
z-CLOSE				0.191** (0.084)	
z-BTWN					0.211** (0.095)
LEVERAGE	-0.178* (0.091)	-0.157* (0.089)	-0.147* (0.089)	-0.155* (0.089)	-0.171* (0.092)
NIM	0.189** (0.076)	0.190** (0.074)	0.195*** (0.072)	0.188** (0.075)	0.200*** (0.076)
SIZE	-0.766 (1.007)	-0.998 (1.023)	-0.951 (1.031)	-1.008 (1.029)	-0.897 (1.028)
EFFICIENCY	0.007 (0.005)	0.006 (0.005)	0.006 (0.005)	0.006 (0.005)	0.007 (0.005)
DEPOSITS	0.890 (1.021)	1.112 (1.033)	1.060 (1.043)	1.131 (1.040)	1.023 (1.041)
PRIVATE	0.790*** (0.199)	0.753*** (0.209)	0.680*** (0.231)	0.798*** (0.198)	0.774*** (0.210)
Constant	1.473 (1.250)	1.527 (1.265)	1.614 (1.328)	1.386 (1.250)	1.401 (1.272)
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	342	342	342	342	342
First stage F-stat	307.756	261.155	113.764	279.152	208.290
Wu-Hausman test (p-val)	0.002	0.009	0.012	0.004	0.000

*p<0.1; **p<0.05; ***p<0.01

Table 5: Board network and volatility of ROA

	<i>Dependent variable: volatility of ROA</i>				
	(1)	(2)	(3)	(4)	(5)
z-DEGREE_1	-0.061** (0.029)				
z-DEGREE_2		-0.064** (0.031)			
z-EIGEN			-0.079** (0.039)		
z-CLOSE				-0.061** (0.029)	
z-BTWN					-0.067** (0.033)
LEVERAGE	0.028 (0.037)	0.021 (0.037)	0.018 (0.037)	0.021 (0.037)	0.026 (0.038)
NIM	-0.022 (0.032)	-0.022 (0.031)	-0.024 (0.031)	-0.021 (0.031)	-0.025 (0.032)
SIZE	0.657* (0.399)	0.730* (0.403)	0.716* (0.406)	0.734* (0.406)	0.699* (0.405)
EFFICIENCY	-0.002 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)	-0.001 (0.002)
DEPOSITS	-0.735* (0.404)	-0.806** (0.407)	-0.789* (0.410)	-0.812** (0.410)	-0.777* (0.410)
PRIVATE	-0.227*** (0.069)	-0.215*** (0.072)	-0.192** (0.079)	-0.229*** (0.069)	-0.221*** (0.072)
Constant	1.662*** (0.424)	1.644*** (0.428)	1.617*** (0.453)	1.689*** (0.424)	1.685*** (0.429)
Year FE	Yes	Yes	Yes	Yes	Yes
Observations	342	342	342	342	342
First stage F-stat	307.756	261.155	113.764	279.152	208.29
Wu-Hausman test (p-val)	0.003	0.011	0.007	0.004	0.000

* p<0.1; ** p<0.05; *** p<0.01

Table 6: Board network and NNPA- Public and private banks

	<i>Dependent variable: NNPA</i>									
	Public sector banks					Private banks				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)
z-DEGREE_1	-0.006** (0.002)					-0.001 (0.001)				
z-DEGREE_2		-0.007** (0.003)					-0.001 (0.001)			
z-EIGEN			-0.011** (0.004)					-0.001 (0.001)		
z-CLOSE				-0.006 (0.501)					-0.001 (0.001)	
z-BTWN					-0.008** (0.003)					-0.001 (0.001)
LEVERAGE	-0.005*** (0.002)	-0.005*** (0.002)	-0.005*** (0.002)	-0.005 (0.445)	-0.005** (0.002)	-0.003 (0.002)	-0.003 (0.002)	-0.003 (0.002)	-0.003 (0.002)	-0.003 (0.002)
NIM	-0.004** (0.002)	-0.004** (0.002)	-0.004** (0.002)	-0.004 (0.472)	-0.004** (0.002)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)	-0.004*** (0.001)
SIZE	0.168*** (0.030)	0.175*** (0.030)	0.180*** (0.031)	0.177 (7.499)	0.178*** (0.031)	0.007 (0.012)	0.009 (0.012)	0.008 (0.012)	0.009 (0.012)	0.008 (0.012)
EFFICIENCY	-0.00002 (0.0001)	0.00003 (0.0001)	0.00005 (0.0001)	0.00003 (0.015)	0.00001 (0.0001)	-0.0002*** (0.0001)	-0.0002*** (0.0001)	-0.0002*** (0.0001)	-0.0002*** (0.0001)	-0.0002*** (0.0001)
DEPOSITS	-0.171*** (0.030)	-0.178*** (0.030)	-0.183*** (0.031)	-0.180 (7.630)	-0.180*** (0.031)	-0.007 (0.012)	-0.009 (0.012)	-0.008 (0.012)	-0.010 (0.012)	-0.008 (0.012)
Constant	0.133*** (0.027)	0.133*** (0.027)	0.130*** (0.027)	0.136 (7.257)	0.128*** (0.028)	0.035** (0.017)	0.036** (0.017)	0.035** (0.017)	0.036** (0.017)	0.037** (0.017)

Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	202	202	202	202	202	138	138	138	138	138
Adjusted R ²	0.808	0.812	0.811	0.815	0.805	0.226	0.230	0.228	0.229	0.227
First stage F-stat	249.072	199.961	93.061	163.510	91.40	131.452	130.408	72.032	153.700	158.033
Wu-Hausman test (p-val)	0.284	0.646	0.353	0.801	0.159	0.545	0.376	0.759	0.354	0.387

*p<0.1; **p<0.05; ***p<0.01

Table 7: Alternate board network variable and GNPA of banks

	<i>Dependent variable: GNPA</i>									
	1 st stage OLS (1)	2 nd Stage IV (2)	1 st stage OLS (3)	2 nd Stage IV (4)	1 st stage OLS (5)	2 nd Stage IV (6)	1 st stage OLS (7)	2 nd Stage IV (8)	1 st stage OLS (9)	2 nd Stage IV (10)
Board size	0.797*** (0.011)		0.936*** (0.108)		1.540*** (0.102)		0.586*** (0.103)		1.194*** (0.105)	
z-DEGREE_1		-0.027*** (0.009)								
z-DEGREE_2				-0.022*** (0.007)						
z-EIGEN						-0.012*** (0.004)				
z-CLOSE								-0.033*** (0.011)		
z-BTWN										-0.016*** (0.005)
PRIVATE	0.848*** (-0.040)	-0.044*** (0.010)	1.067*** (0.126)	-0.041*** (0.011)	1.034*** (0.116)	-0.051*** (0.008)	0.336*** (0.120)	-0.063*** (0.008)	1.288*** (0.119)	-0.041*** (0.010)
LEVERAGE	0.042*** (-0.013)	-0.017*** (0.006)	-0.060 (0.075)	-0.019*** (0.005)	-0.094 (0.062)	-0.022*** (0.005)	-0.014 (0.075)	-0.017*** (0.006)	-0.156** (0.070)	-0.023*** (0.005)
NIM	-0.065*** (-0.013)	-0.015*** (0.003)	-0.058* (0.035)	-0.014*** (0.003)	0.011 (0.037)	-0.013*** (0.003)	-0.103*** (0.036)	-0.015*** (0.003)	-0.043 (0.032)	-0.015*** (0.003)
SIZE	1.174*** (0.396)	0.144*** (0.051)	2.130** (0.989)	0.160*** (0.051)	1.089* (0.576)	0.154*** (0.042)	0.856 (0.804)	0.127** (0.051)	1.926*** (0.740)	0.162*** (0.045)
EFFICIENCY	-0.010***	-0.001**	-0.004	-0.001*	-0.002	-0.001*	-0.010***	-0.001**	-0.001	-0.001*

	(0.0001)	(0.0004)	(0.004)	(0.0004)	(0.006)	(0.0004)	(0.003)	(0.0004)	(0.006)	(0.0004)
DEPOSITS	-0.671	-0.125**	-1.721*	-0.146***	-0.925	-0.147***	-0.409	-0.111**	-1.529**	-0.150***
	(-0.437)	(0.053)	(1.007)	(0.053)	(0.583)	(0.043)	(0.803)	(0.053)	(0.747)	(0.045)
Constant	-8.863***	-0.106	-7.677***	-0.035	-6.423***	0.079	-7.236***	-0.066	-8.279***	0.021
	(0.770)	(0.110)	(0.975)	(0.089)	(0.948)	(0.053)	(0.860)	(0.093)	(0.810)	(0.068)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	342	205	342	205	342	205	342	205	342	205
Adjusted R ²	0.544	0.614	0.538	0.629	0.594	0.687	0.328	0.541	0.629	0.670
First stage F-stat		51.750		66.850		229.1		18.97		132.64
Wu-Hausman test (p-val)		0.000		0.000		0.000		0.001		0.000

*p<0.1; **p<0.05; ***p<0.01